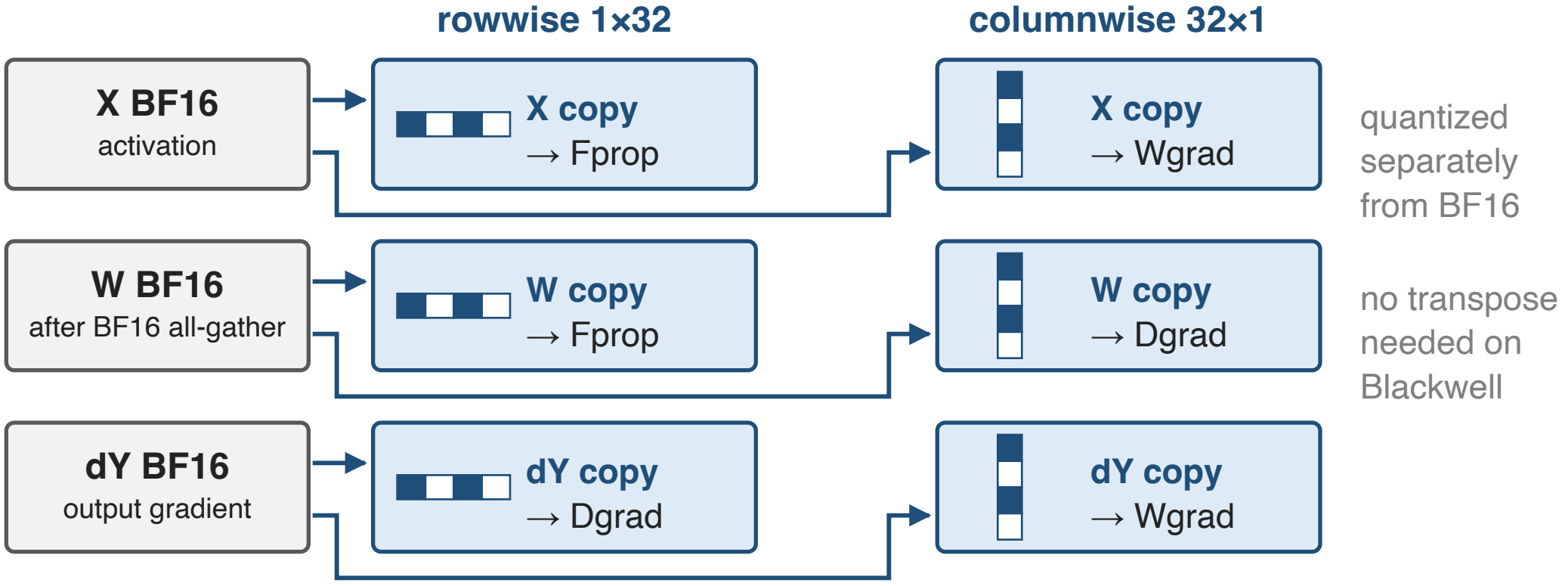
rowwise 1×32
columnwise 32×1
X BF16
activation
X copy
→ Fprop
X copy
→ Wgrad
quantized
separately
from BF16
W BF16
after BF16 all-gather
W copy
→ Fprop
W copy
→ Dgrad
no transpose
needed on
Blackwell
dY BF16
output gradient
dY copy
→ Dgrad
dY copy
→ Wgrad